Anonymous Technical Appendix to Differentially Private All-Pairs Shortest Path Queries on Graphs

Anonymous

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This document is the anonymous technical appendix for the main paper. It contains the full experimental protocol, complete tables and figures, additional proofs, and the appendices omitted from the main paper to meet the page limit. Numbered references to sections, equations, and theorems of the main paper use the main paper's numbering; the appendix sections below use their own A–E numbering.

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This appendix contains additional experimental results, protocol details, and complete proofs. The centralized methods are CSR-SynGraph and GSR-SynGraph, and the mutually exclusive edge-ownership methods are DP–FedGSR, LEG, LDG, and LRG. The Fan and Chen constructions are retained only as explicitly labeled comparison baselines. All topology, route families, spectral bases, ownership partitions, and eigenspace conventions used below are public and independent of private weights.

A Self-Contained Assumptions and Result Statements

The public graph is connected unless a statement is applied componentwise. All edge weights are nonnegative. For a centralized graph, the private database is the edge-weight vector w and neighboring inputs satisfy

Definition A.1 (Weight-private adjacency)**.** Two weight vectors w, w' on the same public topology are adjacent when

$$\|w - w'\|_1 \le 1. \tag{1}$$

For mutually exclusive ownership, $E = \dot{\bigcup}_{i=1}^{K} E_i$ and client i owns w_i on E_i.

Definition A.2 (Bounded one-client adjacency)**.** Federated inputs $\boldsymbol{w}$ and $\boldsymbol{w}'$ are adjacent when one client j may change and

$$w_i = w_i' \ (i \ne j), \qquad \|w_j - w_j'\|_1 \le 1. \tag{2}$$

This protects a bounded change in one client's weight vector; it is not a client-addition or client-replacement guarantee.

A.1 Centralized Guarantees

Let $N = \binom{n}{2}$, let M_r bound the number of public candidate routes for any vertex pair, and let $\tau = \sqrt{2\ln(1.25/\delta)}/\epsilon_{\mathrm{C}}$. For CSR, let $\gamma_r(w)$ be the public-route truncation bias and let $\bar{t}$ be the objective value of the released feasible SDP solution. The route-coupled proof establishes the following bound:

$$\max_{u,v} |\widehat{D}^{\mathrm{CSR}}_{uv} - d_w(u,v)| \le \gamma_r(w) + \tau\sqrt{\bar{t}}\sqrt{2\ln\left(\frac{2NM_r}{\beta}\right)}. \tag{3}$$

When the full chord budget is used, $\gamma_r(w) = 0$; the proof also shows that the full-covariance CSR relaxation is no worse than its diagonal CRH restriction for the same public route workload.

Corollary A.3 (Zero-bias bounded-cycle-rank specialization)**.** *For a public bounded-cycle-rank family with the heavy-light route family, full chord budget, and a ρ-optimal SDP solution, the right-hand side of (3) is bounded by the corresponding heavy-light variance term with $\bar{t} \le L(2(p+1)H_T L + p) + \rho$. In particular, for $p = O(1)$, $\rho = O(\log^3 n)$, and inverse-polynomial failure probability, the order is logarithmic up to the displayed privacy factor.*

For GSR, let U_k be the fixed public line-graph basis, μ_k its coherence, $R_{2,k}$ and $R_{\infty,k}$ the spectral residuals, and H_G the maximum simple-path hop count. With the notation of the main paper, the two valid path comparison bounds are summarized by

$$\max_{u,v} |\widehat{D}^{\mathrm{GSR}}_{uv} - d_w(u,v)| \le \min\{B_{2,k}(w,\beta), B_{\infty,k}(w,\beta)\}. \tag{4}$$

If $\lambda_{k+1} > 0$, spectral smoothness gives $R_{2,k}(w) \le \sqrt{w^{\mathsf{T}} L_{\mathcal{L}} w/\lambda_{k+1}}$ and $R_{\infty,k}(w) \le R_{2,k}(w)$.

Corollary A.4 (Conditional spectral regime)**.** *If $R_{\infty,k}(w) \le \rho_k$, $\mu_k^2 \le C_\mu k/m$, and $m \ge c_m n$, then the ℓ_∞ route bound implies*

$$B_{\infty,k}(w,\beta) \le H_G\left[\rho_k + c_{\epsilon_G,\delta}\mu_k^2\sqrt{2\ln(4m/\beta)}\right]. \tag{5}$$

The bound therefore applies to the stated smoothness/coherence regime.

A.2 Centralized Certificate Statements

For the statements below, let D_T denote the maximum number of tree edges on a public tree path and let L_E denote the number of dyadic levels in the Euler-tour decomposition of the public spanning tree; $L_E = O(\log n)$ in general, and $D_T = O(\log n)$ for a balanced public tree.

Theorem A.5 (Level-aware CSR certificate)**.** *Let the public heavy–light decomposition have dyadic interval coordinates grouped by level ℓ. For a public candidate route P, let $c_{P,\ell}$ be the number of level-ℓ interval coordinates used by its canonical decoder, and define*

$$C_\ell = \max_P c_{P,\ell}, \qquad S_T = \sum_{\ell: C_\ell > 0} \sqrt{C_\ell}, \tag{6}$$

where the maximum ranges over all pairs and all routes in the fixed public family. With the full chord budget $r = p$, the CSR SDP admits a diagonal feasible covariance with maximum route variance at most

$$t_{\mathrm{lev}} = S_T^2 + p. \tag{7}$$

Consequently, for a ρ-optimal numerical solution, with probability at least $1 - \beta$,

$$\begin{aligned} &\max_{u,v} |\widehat{D}_{uv}^{\mathrm{CSR}} - d_w(u,v)| \\ &\quad \le \frac{\sqrt{2\ln(1.25/\delta)}}{\epsilon_{\mathrm{C}}} \sqrt{S_T^2 + p + \rho} \\ &\qquad \cdot \sqrt{2\ln\left(\frac{2NM_p}{\beta}\right)}. \end{aligned} \tag{8}$$

If h_* is the maximum heavy-chain length and H_T the maximum number of heavy chains crossed by a tree path, then $S_T^2 \le 2H_T(\lceil \log_2 h_* \rceil + 1)^2$. For fixed p and inverse-polynomial β, a balanced binary tree satisfies

$$O\left(\frac{\log n \log\log n \sqrt{\ln(1/\delta)}}{\epsilon_{\mathrm{C}}}\right),$$

and a path satisfies

$$O\left(\frac{\log^{3/2} n \sqrt{\ln(1/\delta)}}{\epsilon_{\mathrm{C}}}\right).$$

Theorem A.6 (CSR covariance transfer)**.** *Let $A = A_{\mathrm{C}}$ and let $H \succ 0$ be a feasible precision matrix of a public Gaussian tree-query design, so that $H_{ee} \le 1$ for every edge e. Because the strategy matrix contains singleton rows, A has full column rank. Then there is a feasible CSR covariance Σ with $A^{\mathsf{T}}\Sigma^{-1}A = H$, and the variance of a public route row C_P is $x_P H^{-1} x_P^{\mathsf{T}}$ with $x_P = C_P A$. In particular, taking $H = I$ gives a feasible full covariance with*

$$t^{\mathrm{CSR}} \le (p+1)D_T + p \tag{9}$$

for routes with at most $p+1$ tree segments.

Theorem A.7 (Euler-tour edge-precision certificate)**.** *Let F map the $n-1$ tree-edge weights to the signed depth-first Euler tour of length $2(n-1)$, and let D contain the aligned dyadic interval indicators of this tour, including singletons. The Euler-tour construction yields the public feasible precision matrix $H_{\mathrm{tree}} = B^{\mathsf{T}}B/(4L_E)$ on the tree edges, with chord block I_p, where $B = DF$. It satisfies $H_{ee} \le 1$ for every edge and*

$$t^{\mathrm{CSR}} \le 64(p+1)^2 L_E^2 + p \tag{10}$$

for routes with at most $p+1$ tree segments.

Corollary A.8 (Euler-tour transfer guarantee)**.** *For a ρ-optimal numerical solution of the full CSR program obtained from the Euler-tour precision matrix, with probability at least $1-\beta$,*

$$\begin{aligned}\max_{u,v}|\widehat{D}^{\mathrm{CSR}}_{uv}-d_w(u,v)| \\ \le \frac{\sqrt{2\ln(1.25/\delta)}}{\epsilon_{\mathrm{C}}}\sqrt{64(p+1)^2L_E^2+p+\rho} \\ \cdot\sqrt{2\ln\left(\frac{2NM_p}{\beta}\right)}.\end{aligned} \tag{11}$$

In particular, for fixed p and inverse-polynomial β, $L_E=O(\log n)$ gives $\widetilde{\mathcal{O}}(\log^{3/2} n\sqrt{\log(1/\delta)}/\epsilon_{\mathrm{C}})$.

Theorem A.9 (GSR constant-weight specialization)**.** *Let the public topology be connected with m edges and unweighted hop diameter D_H. On the promised class $w=a\mathbf{1}_m$ for $a\ge 0$, run GSR with $k=1$. Then, with probability at least $1-\beta$,*

$$\max_{u,v}|\widehat{D}^{\mathrm{GSR}}_{uv}-d_w(u,v)|\le\frac{D_H}{m\epsilon_G}\sqrt{2\ln(1.25/\delta)}\sqrt{2\ln(2/\beta)}. \tag{12}$$

For an n-vertex path, $D_H=m=n-1$, so the bound is

$$O\left(\frac{\sqrt{\ln(1/\delta)\ln(1/\beta)}}{\epsilon_G}\right),$$

independent of n.

Theorem A.10 (GSR interval-process bound on paths)**.** *Let the public topology be a path with $m=n-1$ edges and let U_k be the first k line-graph eigenvectors. Assume $\mu_k^2\le C_\mu k/m$ and write $c_{\epsilon,\delta}=\sqrt{2\ln(1.25/\delta)}/\epsilon_G$, $r=(I-U_kU_k^{\mathsf{T}})w$, and*

$$R_{\mathrm{int}}(w)=\max_{0\le i<j\le m}\left|\sum_{e=i+1}^{j}r_e\right|. \tag{13}$$

Suppose the promised weights satisfy

$$\min_e w_e\ge\|r\|_\infty+c_{\epsilon,\delta}\mu_k^2\sqrt{2\ln(4m/\beta)}. \tag{14}$$

Then, with probability at least $1-\beta$, clipping is inactive and

$$\begin{aligned}\max_{u,v}|\widehat{D}^{\mathrm{GSR}}_{uv}-d_w(u,v)| \\ \le R_{\mathrm{int}}(w)+c_{\epsilon,\delta}\sqrt{2C_\mu k\ln\left(\frac{4N}{\beta}\right)},\end{aligned} \tag{15}$$

where $N=\binom{m+1}{2}$.

A.3 Federated Guarantee Statements

For client i, let U_i be its fixed public local basis and μ_i its coherence. The local coefficient query has sensitivity at most μ_i:

Lemma A.11 (DP–FedGSR coefficient sensitivity)**.** *Under Definition A.2,*

$$\|U_i^{\mathsf{T}}w_i-U_i^{\mathsf{T}}w_i'\|_2\le\mu_i.$$

Consequently, with Gaussian scale calibrated to (ϵ_i,δ_i), the joint transcript and its APSP post-processing satisfy

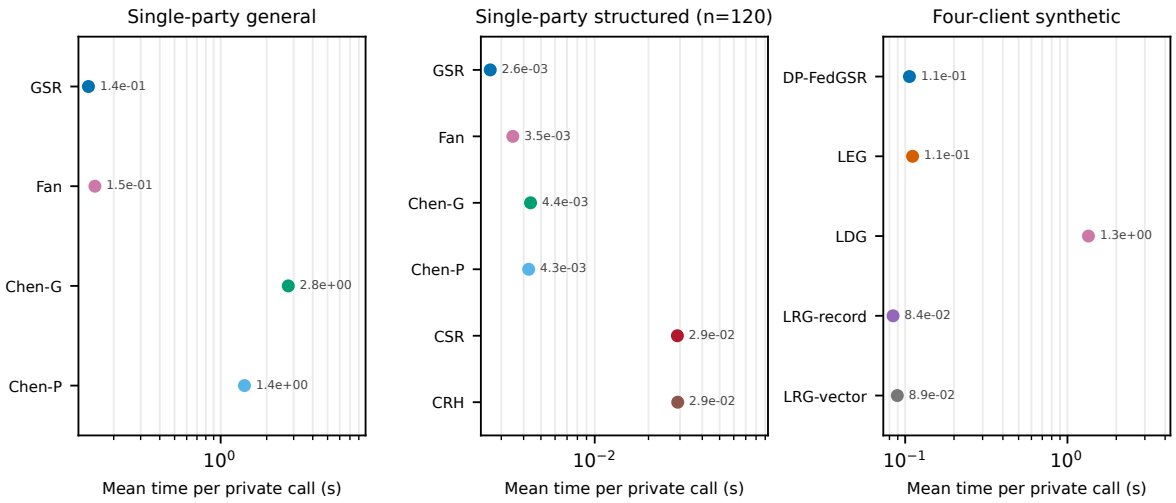


Figure 1: Mean elapsed time of one private method call at $\epsilon = 1$. The left panel uses the 54 grid/ER/BA graphs for single-party methods; the middle panel uses the 12-graph $n = 120$ tree/grid block, the only completed block that contains CSR, CRH, and the Fan/Chen/GSR comparators, so it is not a general-graph runtime; the right panel uses 108 six-topology graphs for four-client methods. Each graph is weighted equally after averaging its 20 noise repeats. The horizontal axes are logarithmic; times exclude loading, the nonprivate oracle, process scheduling, and one-time public design precomputation (the GSR basis and the CSR full-covariance SDP, the latter taking a median 9.5 min per $n = 120$ graph and cached across budgets and noise repeats; the CRH rows in this block use the isotropic variant, which needs no SDP). Historical runs do not have a uniform hardware log, so these are descriptive timings, not a hardware-normalized throughput comparison.

Theorem A.12 (DP–FedGSR privacy)**.** $(\epsilon_{\max}, \delta_{\max})$*-DP for the bounded one-client adjacency, where* $\epsilon_{\max} = \max_i \epsilon_i$ *and* $\delta_{\max} = \max_i \delta_i$.

Let $B_i(\beta)$ denote the local residual-plus-noise envelope defined by the local reconstruction in the main paper. Then

Theorem A.13 (DP–FedGSR edge utility)**.** *With probability at least* $1 - \beta$*, every edge owned by client* i *has absolute release error at most* $B_i(\beta)$.

Theorem A.14 (DP–FedGSR APSP utility)**.** *With the same probability,*

$$\max_{u,v} |\widehat{D}_{uv} - D_{uv}| \le H_G \max_i B_i(\beta). \tag{16}$$

B Additional Experimental Results and Protocol Details

The single-party delta sweep at $\epsilon = 1$ and $n = 480$ produces GSR MAE 9.02, 8.93, and 8.86 for $\delta = 10^{-3}, 10^{-5}, 10^{-7}$, respectively. For the corresponding four-client block, DP-FedGSR has MAE 31.82, 31.92, and 31.99. These are separate parameter blocks, not observations from the main privacy sweep; they expose the sensitivity of the spectral reconstruction to the Gaussian failure-probability parameter.

The tree/grid baselines are evaluated only on their applicable generated topologies. The Fan and Li tree/grid methods [1] and the Zou et al. generalized binary-tree grid method [2] use the completed 20-repeat specialized blocks. The earlier 10-repeat CSVs are excluded from all active tables and figures.

B.1 Single-Party Breakdowns and Sensitivity

The rank sweep is a diagnostic rather than a retrospective parameter choice. Single-party GSR changes from MAE 7.55 at $k = 32$ to 7.40 at $k = 128$; DP-FedGSR changes from 39.85 to 10.45. The observed MAE

Table 1: Ablation details at $\epsilon = 1$ (MAE, lower is better). The CSR/CRH row uses the 12-graph $n = 120$ tree/grid block with 20 noise repeats; the other rows use the same 36 synthetic graphs at $n = 480$ with 10 noise repeats and the prespecified rank $k = 32$.

Setting	GSR/CSR	Comparator
Centralized, smooth	4.15	15.35 (identity)
Centralized, random	9.34	11.72 (identity)
Four clients, smooth	37.59	15.15 (edge Gaussian)
Four clients, random	34.81	11.64 (edge Gaussian)
Structured ($n = 120$) CSR vs CRH	23.58	29.74 (CRH)

Table 2: Ownership ablation at $n = 480$, $\epsilon = 1$, four clients, 10 noise repeats. MAE is averaged over 24 synthetic graph instances per partition scheme.

Partition	DP-FedGSR	Edge Gaussian
Uniform	25.86	13.14
Skewed	20.55	13.13
Bridge-stress	14.86	13.03

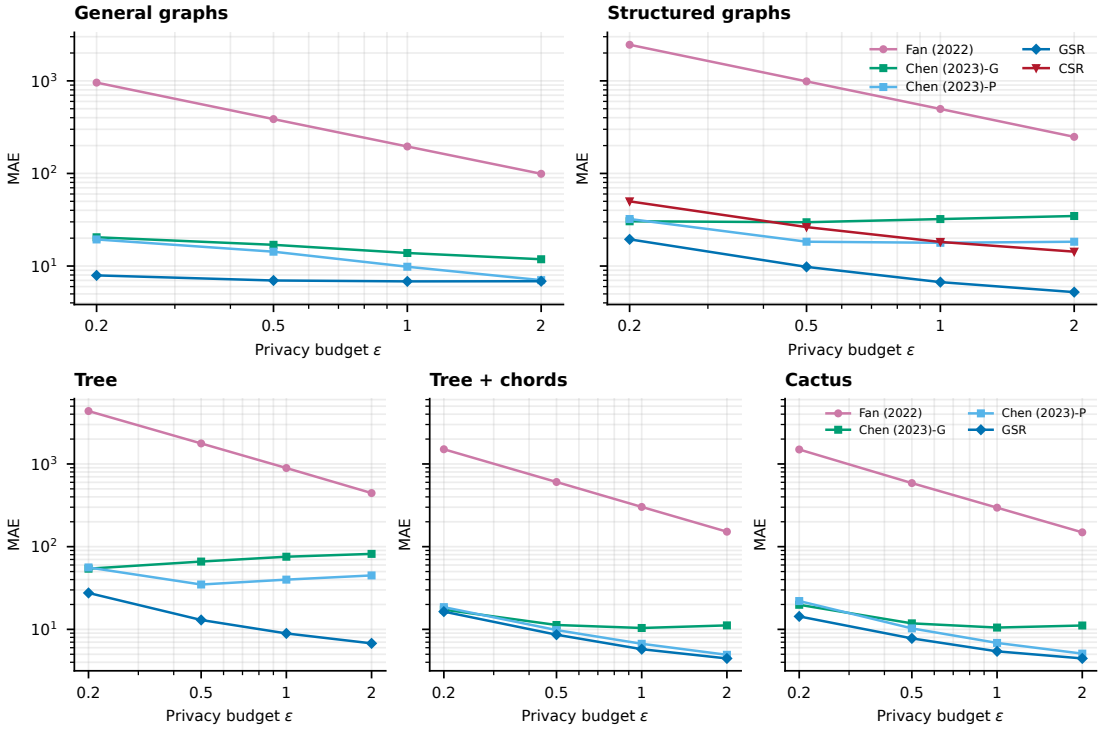


Figure 2: Detailed single-party privacy and topology breakdowns. Points are graph-macro means and shaded regions are 95% graph-level confidence intervals. CSR appears only in the structured panel: its route construction requires a tree-with-chords backbone and is not defined on the grid/ER/BA graphs in the left panel.

values for $K = 2, 4, 8, 16$ clients are 25.73, 38.04, 33.24, and 19.59.

At $\epsilon = 1$, replacing the full-covariance CSR design by the diagonal/isotropic CRH release increases graph-macro MAE from 13.15 to 20.45 at $n = 120$, from 16.04 to 23.96 at $n = 240$, from 18.95 to 27.64 at $n = 480$, and from 22.56 to 32.54 at $n = 960$. The CRH/CSR ratio ranges from 1.44 to 1.56 at $\epsilon = 1$, from 1.65 to

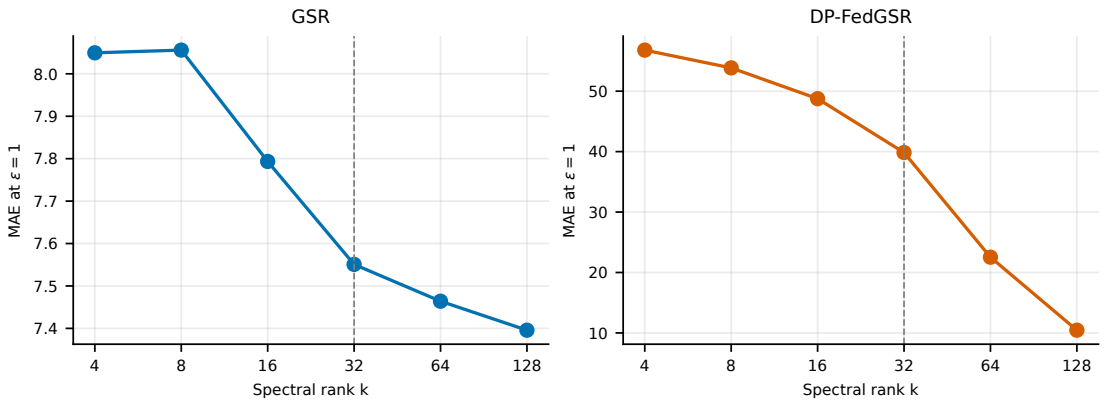


Figure 3: Sensitivity to the public spectral rank k at $\epsilon = 1$. The dashed reference is the prespecified main choice $k = 32$.

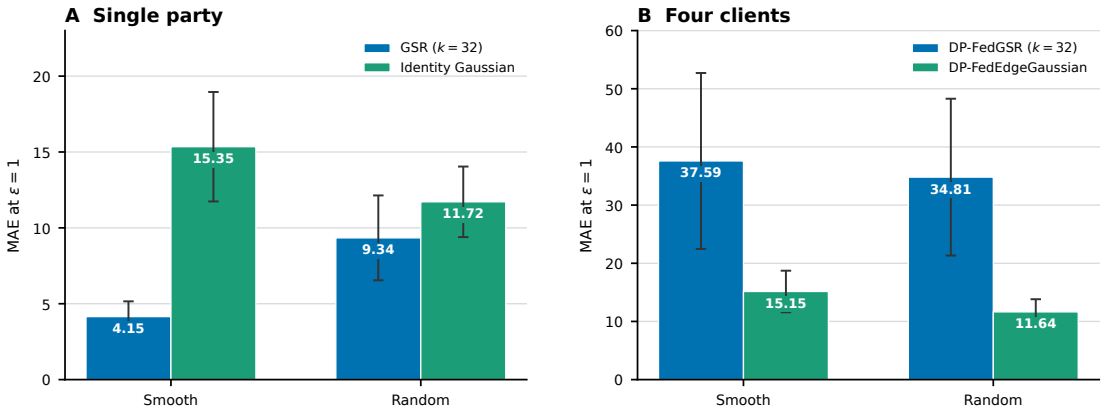


Figure 4: Identity-basis ablation at $n = 480$, $\epsilon = 1$, and $k = 32$: centralized GSR versus an edge-wise identity-basis Gaussian release (left), and four-client DP–FedGSR versus DP–FedEdgeGaussian (right). Each panel uses the same 36 synthetic graphs (six topologies, smooth/random weights, three seeds); each bar averages the 18 graphs of its weight family with 10 noise repeats per graph, and whiskers are 95% graph-level confidence intervals. The learned centralized basis helps on both weight families, whereas the local per-client bases at $k = 32$ do not; the rank sweep in Fig. 3 shows that $k = 128$ closes the gap.

1.77 at $\epsilon = 0.2$, and from 1.26 to 1.36 at $\epsilon = 2$. The separate eight-method $n = 120$ structured block reports MAE 23.58 versus 29.74 for CSR and CRH, respectively.

On the structured block, CSR improves over CRH at $\epsilon = 1$ (MAE 23.58 versus 29.74).

At $\epsilon = 1$, NMAE is 0.297, 0.292, 0.304, and 0.308 at $n = 120, 240, 480, 960$, respectively, so the relative accuracy is essentially unchanged by an eightfold increase in graph size. MAE grows from 13.15 to 16.04, 18.95, and 22.56 as the distance scale increases. In the matched comparisons, GSR is more accurate at every size; CSR is comparable to Chen-P at $n = 480$ and $n = 960$ and substantially better than Fan. The dense route matrix becomes the memory bottleneck at these sizes, so the scale block stores each route quadratic form as a sparse row over the covariance entries. This reformulation is mathematically identical to the dense SDP (matching objectives at $n = 120$ and $n = 480$) and keeps a single $n = 960$ solve near 8–15 GB.

B.2 Scalability and Runtime

Graph-macro mean GSR call time grows from 0.41 s at $n = 960$ to 25.60 s at $n = 7680$; DP-FedGSR grows from 0.43 s to 58.76 s. The Chen-style Gaussian call has a graph-macro mean of 166 s at $n = 7680$ (per-graph median 162 s). Peak RSS and stage-separated timings were not recorded consistently, so no memory

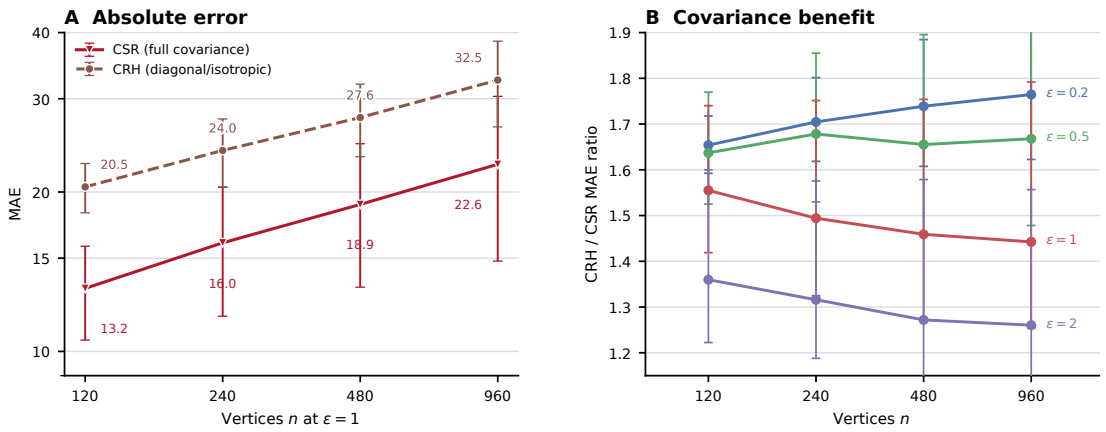


Figure 5: CSR covariance ablation across graph size. Both mechanisms use the same 18 structured graphs per size (tree, tree-with-chords, and cactus; smooth and random weights; three seeds), the same tree-path route workload ($r = 0$), and 20 noise repeats, so the comparison isolates the covariance design. Panel A shows graph-macro MAE at $\epsilon = 1$ with 95% graph-level confidence intervals; panel B shows the CRH/CSR ratio of graph-macro means for each budget with paired bootstrap 95% confidence intervals. The full-chord-budget $n = 30$ ablation is a separate block.

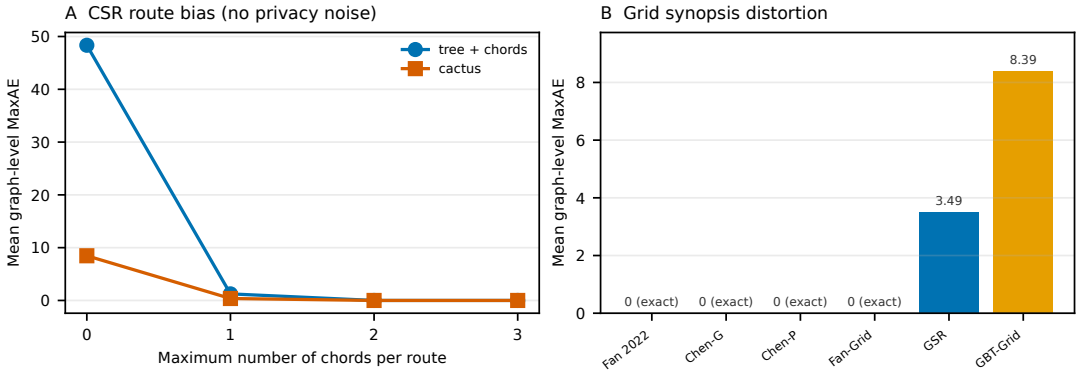


Figure 6: Zero-noise structural diagnostic on the $n = 30$ structured block. Panel A reports the route-restricted CSR bias as the chord budget grows; at zero noise CRH shares the same route family and coincides with CSR, so the difference between the two private mechanisms is covariance design, not route coverage. Panel B reports the deterministic structural distortion of each executable synopsis on grid graphs with privacy noise disabled; Fan and Chen reconstruct grid distances exactly, whereas the truncated spectral and binary-tree synopses retain a nonzero bias. This is a mechanism check, not a private-utility comparison.

or hardware-normalized speed claim is made.

Additional result files retain RMSE, 95th-percentile absolute error, NRMSE, and private method-call time in addition to the metrics shown in the paper. Source-level real-network NMAE and graph-level confidence intervals are generated from the same fixed graph collection and evaluation protocol.

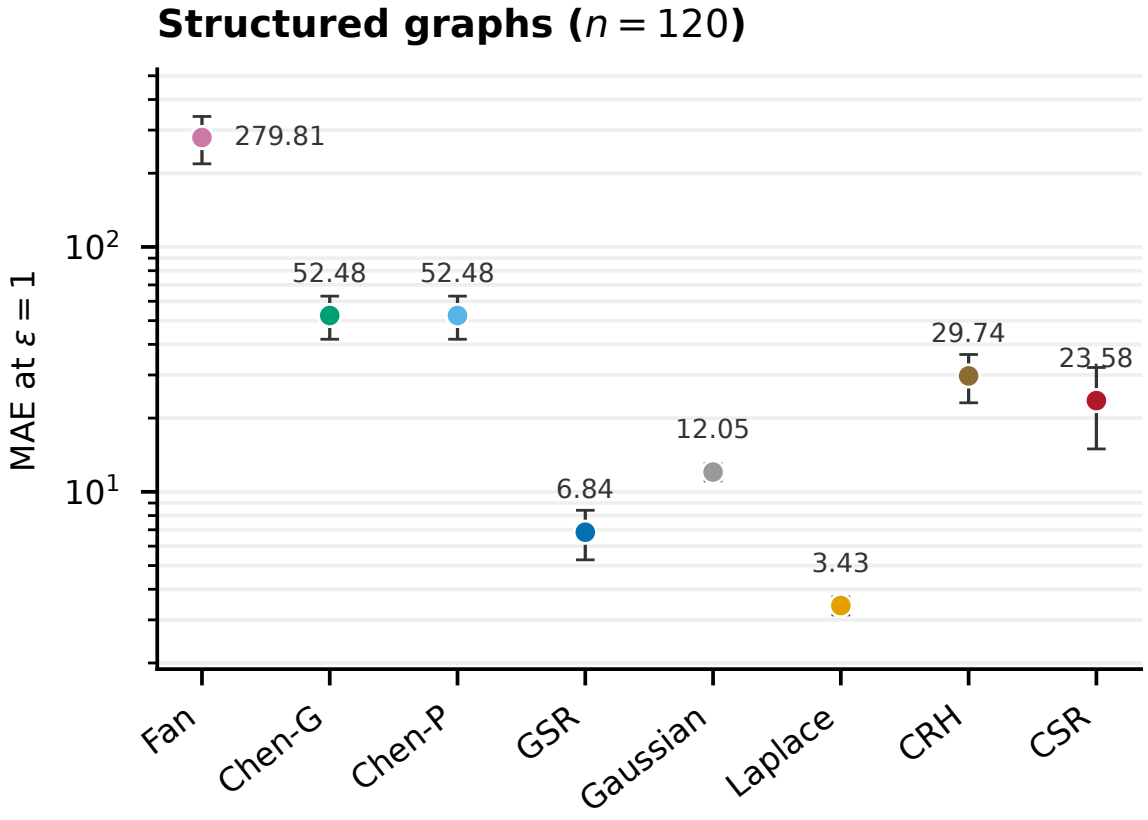


Figure 7: Structured-graph CSR/CRH comparison at $n = 120$ and $\epsilon = 1$. This is the complete eight-method comparator block on the same 12 tree/grid graphs, including CRH, identity-basis Gaussian, and Laplace; the larger-n CSR scale diagnostic is reported in Fig. 8C, where matched baselines exist at $n = 120, 480, 960$.

Table 3: CSR scale diagnostic on structured graphs at $n \in \{120, 240, 480, 960\}$. Each cell is a graph-macro average over 18 graphs (tree, tree-with-chords, and cactus; smooth and random weights; three seeds) and 20 noise repeats. All 5,760 releases solve to optimal or optimal-inaccurate status, and no release is all-zero. The scale block uses the $r = 0$ tree-path workload; the full-chord zero-bias guarantee is covered by the separate $n = 30$ ablation, and the complete eight-method CSR/CRH block uses an $n = 120$ tree/grid protocol that is not directly comparable.

Size	$\epsilon = 0.2$	$\epsilon = 0.5$	$\epsilon = 1$	$\epsilon = 2$
MAE				
$n = 120$	40.45	20.60	13.15	9.54
$n = 240$	46.15	23.85	16.04	12.17
$n = 480$	51.65	27.31	18.95	15.01
$n = 960$	57.84	31.12	22.56	18.34
NMAE				
$n = 120$	0.873	0.454	0.297	0.222
$n = 240$	0.776	0.418	0.292	0.231
$n = 480$	0.747	0.416	0.304	0.252
$n = 960$	0.694	0.399	0.308	0.264

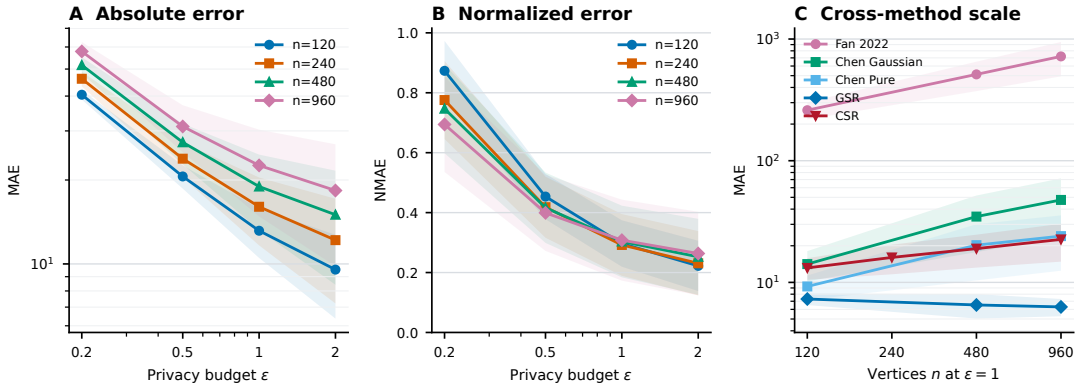


Figure 8: CSR scale diagnostic at $n \in \{120, 240, 480, 960\}$. Points are graph-macro means over 18 structured graphs per size; ribbons are 95% confidence intervals across graphs. Panel A reports MAE and panel B reports the scale-normalized NMAE. Panel C places CSR at $\epsilon = 1$ next to the Fan, Chen, and GSR rows evaluated on exactly the same graphs; the baselines exist at $n = 120, 480, 960$, while $n = 240$ is CSR-only. The normalized curves are close at $\epsilon \in \{1, 2\}$, whereas absolute error grows with the distance scale of the larger graphs. All 5,760 CSR releases solve to optimal or optimal-inaccurate status, and no release is all-zero.

C Full Experimental Evaluation

C.1 Questions, Protocol, and Reproducibility

We ask whether graph-spectral reconstruction improves weight-private APSP utility, for which topology and signal classes, and whether the benefit survives mutually exclusive edge ownership. We also test CSR on small structured graphs.

The topology and vertex identities are public; edge weights are private with $\|w - w'\|_1 \le 1$. The main privacy sweep is $\epsilon \in \{0.2, 0.5, 1, 2\}$; Gaussian mechanisms use $\delta = 10^{-5}$, and pure-DP mechanisms $\delta = 0$. Classical Gaussian calibration is stated for $\epsilon \le 1$, so the $\epsilon = 2$ rows are an empirical sensitivity point outside those theorem guarantees. GSR and DP-FedGSR use prespecified $k = 32$. Main blocks use 20 noise repeats per graph/epsilon/method, parameter and ablation blocks 10, and scalability 3. The multi-party block uses four clients and a uniform exclusive edge partition. Runtime is wall time inside the method call, excluding graph loading and nonprivate reference computation.

The 252 synthetic graphs cover tree, tree-with-chords, cactus, grid, Erdos–Renyi (ER), and Barabasi–Albert (BA) topologies, seven sizes from 30 to 1920 vertices, smooth and random weights, and three topology seeds. The main centralized comparison uses 54 general graphs (grid/ER/BA) and 54 structured graphs (tree/tree-with-chords/cactus) at $n \in \{120, 480, 960\}$; the multi-party comparison uses all 108. Real data contain 159 sampled graphs from 12 sources, with 136 in the fixed main filter. Units differ across sources, so absolute errors are never pooled across sources.

The general-graph comparators reimplement Fan *et al.* [3] and the Gaussian and pure-DP Chen *et al.* [4] constructions under the common weight-private model and matched privacy parameters. The tree and grid variants follow Fan and Li [1], and the generalized binary-tree implementation follows Zou *et al.* [2]. The low-tree-width algorithm [5] is omitted because no validated public tree-decomposition oracle is available.

C.2 Metrics and Statistical Unit

Let $\mathcal{Q} = \{\{u, v\} : u < v\}$ on a connected graph, let $a_{uv} = |\widehat{d}(u, v) - d_w(u, v)|$, and let $\bar{d} = |\mathcal{Q}|^{-1} \sum_{\mathcal{Q}} d_w(u, v)$. We report

$$\begin{aligned} \mathrm{MAE} &= |\mathcal{Q}|^{-1} \sum_{\mathcal{Q}} a_{uv}, \\ \mathrm{MaxAE} &= \max_{\mathcal{Q}} a_{uv}, \\ \mathrm{RMSE} &= \left(|\mathcal{Q}|^{-1} \sum_{\mathcal{Q}} (\widehat{d} - d_w)^2 \right)^{1/2}, \\ \mathrm{NMAE} &= \mathrm{MAE}/\bar{d}. \end{aligned} \tag{17}$$

Each table cell first averages noise repeats on a graph, then averages graphs equally. Confidence intervals use variation across graph-instance means; noise repeats of one graph are not treated as independent networks. The nonprivate oracle defines target distances and is excluded from private-method averages.

C.3 Computational and Communication Complexity

Let $n = |V|$, $m = |E|$, and let $T_{\mathrm{APSP}}(n, m)$ denote the cost of the nonnegative APSP routine used after reconstruction. For sparse Dijkstra, $T_{\mathrm{APSP}}(n, m) = O(nm + n^2 \log n)$. With public spectral rank k, GSR has time $O(T_{\mathrm{basis}}(m, k) + mk + T_{\mathrm{APSP}}(n, m))$ and space $O(m + |E_{\mathcal{L}}| + mk + n^2)$, where T_{basis} is the truncated line-graph eigensolver cost. If d is the tree–chord coordinate dimension and q is the CSR route-query count, CSR has time $O(T_{\mathrm{SDP}}(d, q) + qd + d^2 + T_{\mathrm{APSP}}(n, m))$ and space $O(m + qd + d^2 + n^2)$. CRH is the diagonal restriction, with time $O(qd + T_{\mathrm{APSP}}(n, m))$ and space $O(m + qd + n^2)$.

For K mutually exclusive clients, client i has (n_i, m_i), spectral rank k_i, and q_i local shortcuts. DP–

Table 4: Centralized synthetic APSP error at $\epsilon = 1$ with $\delta = 10^{-5}$ for approximate-DP methods and $\delta = 0$ for pure DP. Panel (a) averages 54 general graphs and (b) 54 structured graphs at $n \in \{120, 480, 960\}$ with 20 repeats; lower is better. CSR is evaluated only in panel (b), with the $r = 0$ tree-path workload; its tree-chord workload is intractable on general graphs. Chen rows use the repaired bounded-hop composition described in the text.

(a) General graphs: grid/ER/BA

Method	MAE	MaxAE
Fan et al. (2022) reimplementation [3]	195.57	497.03
Chen et al. (2023) Gaussian reimplementation [4]	13.83	54.62
Chen et al. (2023) pure-DP reimplementation [4]	9.83	45.24
GSR-SynGraph ($k = 32$)	6.83	26.24

(b) Structured graphs: tree/tree-with-chords/cactus

Method	MAE	MaxAE
Fan et al. (2022) reimplementation [3]	497.36	2286.31
Chen et al. (2023) Gaussian reimplementation [4]	32.16	155.84
Chen et al. (2023) pure-DP reimplementation [4]	17.83	123.85
GSR-SynGraph ($k = 32$)	6.70	34.01
CSR-SynGraph ($r = 0$)	18.22	86.54

FedGSR takes

$$O\Big(\sum_i [T_{\text{basis}}(m_i, k_i) + m_i k_i] + T_{\text{APSP}}(n, m) \Big)$$

time, uses $O(m_i + |E_{\mathcal{L}_i}| + m_i k_i)$ client space and $O(m + n^2)$ server space, and communicates $O(\sum_i k_i)$ real coefficients. LEG and LRG take $O(\sum_i m_i + T_{\text{APSP}}(n, m))$ time, use $O(m_i)$ client and $O(m + n^2)$ server space, and communicate $O(\sum_i m_i) = O(m)$ records. LDG takes $O(\sum_i T_{\text{APSP}}(n_i, m_i) + T_{\text{APSP}}(n, \sum_i q_i))$ time, communicates $O(\sum_i q_i)$ shortcuts (at most $O(\sum_i n_i^2)$), and uses $O(n_i^2)$ client space and $O(n^2 + \sum_i q_i)$ server space. These are payload counts and exclude network protocol headers.

C.4 Main Centralized Results

Detailed single-party breakdown panels are in the appendix.

At $\epsilon = 1$, GSR has lower MAE than Fan and the repaired Chen implementations on both groups (6.83 versus 195.57, 13.83, and 9.83 on general graphs; 6.70 versus 497.36, 32.16, and 17.83 on structured graphs). Chen-pure outperforms Chen-Gaussian through its pure-DP calibration. The Chen rows treat negative noisy estimates as missing ($+\infty$) instead of clipping to zero; this post-processing convention is applied uniformly.

CSR is evaluated only on structured graphs, where it attains MAE 18.22 and MaxAE 86.54 at $\epsilon = 1$ with $r = 0$; GSR is stronger on both metrics. The general graphs have $p = \Theta(n)$ (about 900 for a 960-vertex grid and 12,300 for ER), which makes $r \geq 1$ intractable; the reported block is therefore an end-to-end tree-path release.

In the topology-specialized diagnostics at $\epsilon = 1$, the Fan heavy–light adaptation has MAE 15.82 over 18 tree instances and the Fan grid-block adaptation 1523.49 over 18 grid instances. These are separate from the 54-instance ranking. The Zou et al. method is reported on grids, where its structural assumptions apply.

C.5 Real Networks and Source-Level Reporting

The real-network analysis uses the completed 136-graph, 20-repeat blocks for Fan, Chen-Gaussian, Chen-pure, and GSR. Because physical units differ, NMAE is averaged within each graph and then equally across

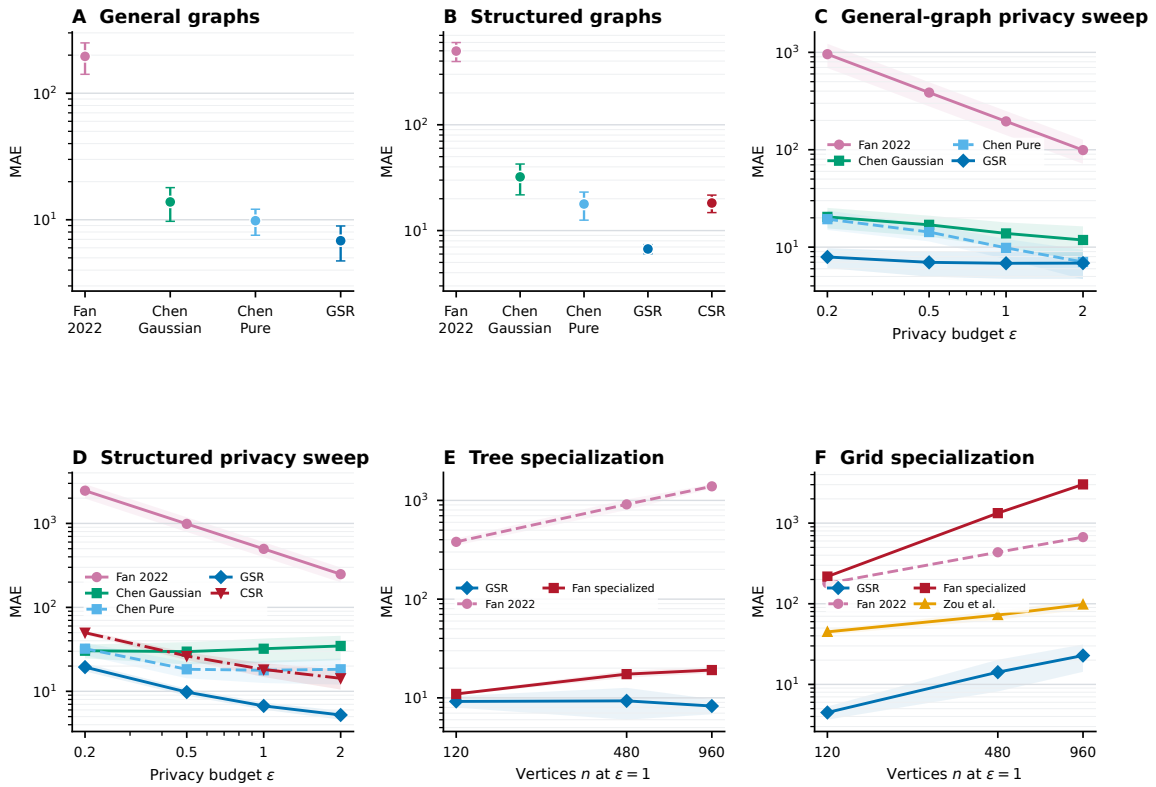


Figure 9: Single-party study. (A,B) Recent Fan/Chen implementations and GSR at $\epsilon = 1$, with CSR added in (B); error bars are 95% graph-level confidence intervals. (C,D) Privacy sweeps. CSR appears only where its tree-chord workload is tractable. (E,F) Tree/grid specializations, including the Zou et al. generalized binary-tree grid method [2]. The $\epsilon = 2$ points lie outside the classical Gaussian guarantee.

graphs from the same source; values above one are retained. Source-level curves are in the appendix.

C.6 Federated Methods and Results

LEG, LDG, and LRG are our methods for mutually exclusive edge ownership, with different privacy units: LEG releases perturbed owned edges, LDG releases local distance shortcuts, and LRG applies bounded local randomization to edge records. DP-FedGSR is the spectral method in the same family. Fan/Chen local adaptations run the corresponding centralized mechanism on each client's public component and unite the released shortcuts.

All multi-party methods use one data-dependent client-to-server round: each client sends one privatized synopsis or record set, and the server reconstructs, unions, and computes APSP. The Fan/Chen rows use the same one-round structure.

At $\epsilon = 1$, LEG has the lowest MAE among the proposed edge/distance methods in Table 5 (3.55), compared with 34.88 for DP-FedGSR. The gap reflects the released objects: LEG preserves every edge at an m-coordinate payload, whereas DP-FedGSR discards local high-frequency residuals. The DP-FedGSR curve flattens once coefficient noise falls below this residual. LDG and both LRG variants are less accurate.

For real networks at $\epsilon = 1$, LEG attains NMAE 0.438, versus 4.49 for DP-FedGSR, 6.96 and 8.30 for Chen-pure and Chen-Gaussian, and 6842.26 for Fan; LDG (22.34) and both LRG units are higher.

On synthetic graphs, Fan/Chen local adaptations are much weaker than LEG (MAE 25,820.23, 597.57, and 509.56 versus 3.55). They apply the centralized mechanisms per client with valid per-component budgets and test how recent centralized ideas transfer to exclusive ownership.

The remaining ownership, delta, client-count, real-network, and scalability analyses are in the appendix.

Table 5: Four-client synthetic comparison at $\epsilon = 1$ with uniform exclusive edge ownership: 108 graphs, 20 noise repeats. Runtime is descriptive and excludes loading and oracle computation. Pure-DP and local-DP rows use $\delta = 0$; Gaussian rows use $\delta = 10^{-5}$. LRG–Vector protects a different privacy unit and is not a matched-privacy comparison. Fan/Chen rows use per-client component budgets and the repaired Chen composition.

Method	Role / privacy unit	MAE	MaxAE	Runtime (s)
DP-FedGSR ($k = 32$)	Proposed; edge vector, approx. DP	34.88	97.62	0.106
LEG	Proposed; edge vector, pure DP	3.55	18.49	0.111
LDG	Proposed; local distance vector, approx. DP	1346.75	6687.26	1.347
LRG–PerRecord	Proposed; one edge record, local DP	58.71	209.12	0.084
LRG–Vector	Proposed; one client vector, local DP	12274.94	38010.96	0.089
FedFan–Local	Baseline adaptation	25820.23	86667.58	1.432
FedChen–Gaussian–Local	Baseline adaptation	597.57	2259.21	3.718
FedChen–Pure–Local	Baseline adaptation	509.56	1988.21	3.644

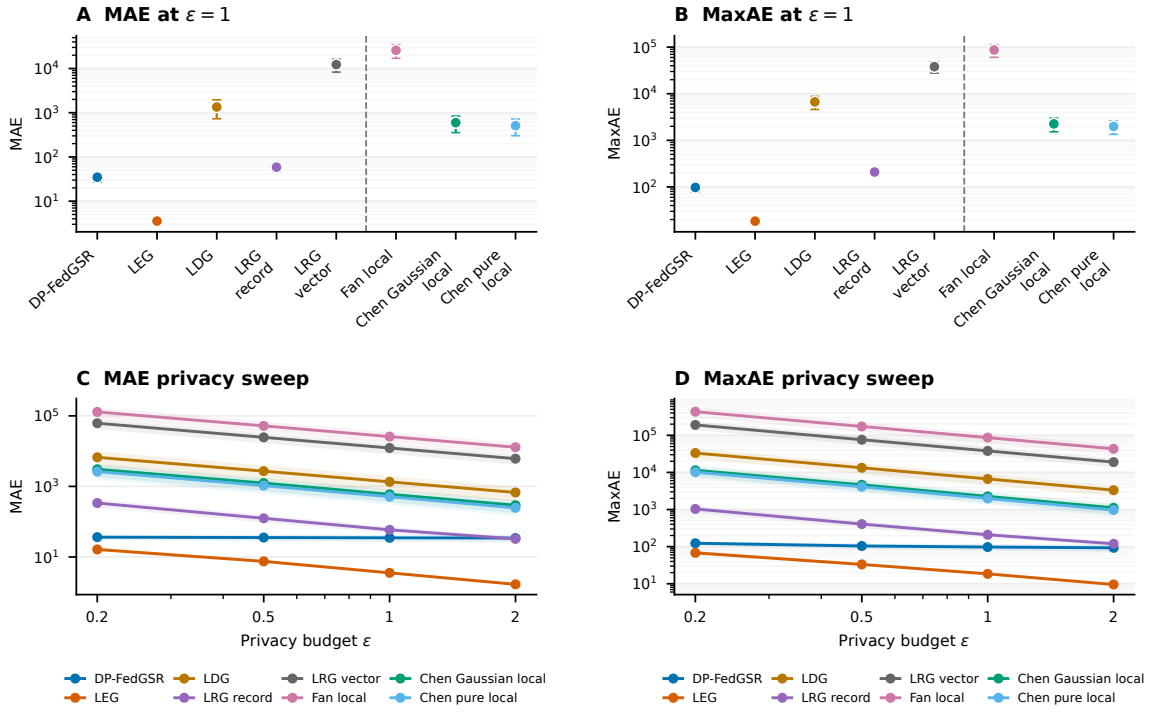


Figure 10: Multi-party comparison under four-client uniform exclusive edge ownership. (A,B) MAE and MaxAE at $\epsilon = 1$; (C,D) privacy-budget sweeps. Points are graph-macro means over 108 graphs and 20 repeats; error bars are 95% graph-level confidence intervals.

C.7 Design Diagnostics

Table **??** collects the covariance, identity-basis, and public-rank diagnostics summarized in the main paper. Panel (a) replaces the full-covariance CSR precision matrix by its diagonal CRH restriction on the same tree-path workload ($r = 0$), so the gap measures covariance design rather than route coverage; the CRH/CSR ratio stays between 1.44 and 1.56, and the nesting argument used for the certificate comparison explains why the full-covariance relaxation cannot be worse. Panel (b) replaces the learned line-graph basis by an edge-wise identity basis: centralized GSR wins on both weight families, whereas at the fixed local rank $k = 32$ DP-FedGSR loses to edge-wise Gaussian. Panel (c) shows that raising the local rank to $k = 128$ eliminates that gap (10.45 versus 15.15 and 11.64) while centralized GSR barely moves, so local rank rather

Table 6: Design diagnostics at $\epsilon = 1$ (MAE; lower is better). Panel (a): 18 structured graphs per size, 20 repeats; the ratio is CRH/CSR, so a value above one means CRH is worse. Panels (b,c): 36 and 24 graphs at $n = 480$, 10 repeats. In (b), "Spectral" is GSR (centralized) or DP-FedGSR (four clients), and "Edge Gaussian" is edge-wise Gaussian release. In (c), k is the global rank for GSR and the per-client local rank for DP-FedGSR.

(a) CSR covariance, 18 graphs per size

n	CSR	CRH	CRH/CSR
120	13.15	20.45	1.56
240	16.04	23.96	1.49
480	18.95	27.64	1.46
960	22.56	32.54	1.44

(b) Identity basis, $n = 480$ (36 graphs)

Setting	Weights	Spectral	Edge Gaussian
Centralized	Smooth	4.15	15.35
Centralized	Random	9.34	11.72
Four clients	Smooth	37.59	15.15
Four clients	Random	34.81	11.64

(c) Public rank, $n = 480$ (24 graphs)

k	GSR	DP-FedGSR
32	7.55	39.85
128	7.40	10.45

than spectral representation per se governs the federated accuracy–communication frontier. CSR scale diagnostics additionally give NMAE 0.292–0.308 for $n = 120, \ldots, 960$ at $\epsilon = 1$.

D Theoretical Proofs

This appendix proves the active theoretical statements in Sections III and IV. All public bases, route families, ownership partitions, and eigenspace basis choices are fixed independently of the private weights. If a requested prefix cuts through a repeated eigenspace, the selected deterministic basis is part of the public preprocessing; when a complete repeated eigenspace is included, its projector is invariant to basis rotations. The proofs below hold for the selected public basis. They use the weight-private adjacency relation in Definition A.1 and the bounded one-client relation in Definition A.2.

D.1 CRH-Opt Algorithm and Guarantee

The complete CRH-Opt procedure and its formal guarantee are stated here so that the main paper can focus on the public covariance design. The public program and release equations are given in Section 3.2.1.

Algorithm 1 CRH-Opt

Input: Public topology, public spanning tree T, private weights w, chord budget r, ϵ_{C}, and δ
Output: Symmetric nonnegative APSP estimate matrix $\widehat{D}^{\mathrm{C}}$

1: Construct A_{C}, the public route families $\mathcal{P}_r(u, v)$, and rows C_P
2: Solve the public convex program (13)
3: Release $y \leftarrow A_{\mathrm{C}} w + \tau D_* g$ using (14) and (15)
4: For every pair, compute $\widehat{D}^{\mathrm{C}}_{uv} \leftarrow \max\{0, \min_{P \in \mathcal{P}_r(u,v)} C_P y\}$
5: Set the diagonal to zero, symmetrize, and return

Theorem D.1 (CRH-Opt Privacy and Uniform Error)**.** *For $0 < \epsilon_{\mathrm{C}} \leq 1$ and $0 < \delta < 1$, Algorithm 1 satisfies*

$(\epsilon_{\mathrm{C}}, \delta)$-*differential privacy. Let* $N = \binom{n}{2}$ *and* $\beta \in (0,1)$. *With probability at least* $1-\beta$,

$$\max_{u,v\in V} |\widehat{D}^{\mathrm{C}}_{uv} - d_w(u,v)| \le \gamma_r(w) + \frac{\sqrt{2\ln(1.25/\delta)}}{\epsilon_{\mathrm{C}}}\sqrt{t^*_r} \cdot \sqrt{2\ln\left(\frac{2NM_r}{\beta}\right)}. \tag{18}$$

Moreover, the isotropic choice $\lambda_i = \Delta^2_{\mathrm{C}}$ *is feasible in* (13). *Consequently,*

$$t^*_r \le \Delta^2_{\mathrm{C}} b^2_{\mathrm{C},r}, \tag{19}$$

so CRH-Opt's certified noise term is never larger than that of the isotropic coordinate release, and is strictly smaller whenever the inequality is strict. For $r = p$, $\gamma_p(w) = 0$.

D.2 Proof of CRH-Opt

Let $A = A_{\mathrm{C}}$ and let $D_* = \mathrm{diag}(\sqrt{\lambda^*_1}, \dots, \sqrt{\lambda^*_q})$. For adjacent w, w', put $x = w - w'$. The public sensitivity constraints give

$$\|D_*^{-1} A_{:,e}\|_2^2 = \sum_i A^2_{ie}/\lambda^*_i \le 1.$$

Consequently,

$$\|D_*^{-1}Aw - D_*^{-1}Aw'\|_2 \le \sum_e |x_e| \|D_*^{-1}A_{:,e}\|_2 \le \|x\|_1 \le 1.$$

The whitened query therefore has ℓ_2 sensitivity at most one. The standard Gaussian mechanism with scale $\sqrt{2\ln(1.25/\delta)}/\epsilon_{\mathrm{C}}$ is $(\epsilon_{\mathrm{C}}, \delta)$-DP for $0 < \epsilon_{\mathrm{C}} \le 1$. Multiplication by the public matrix D_* and route reconstruction are post-processing.

For a candidate route P, write

$$C_P y = w(P) + \zeta_P, \qquad \zeta_P = \tau C_P D_* g, \qquad \tau = \frac{\sqrt{2\ln(1.25/\delta)}}{\epsilon_{\mathrm{C}}}.$$

The route constraint gives $\mathrm{Var}(\zeta_P) = \tau^2 \sum_i C^2_{P,i}\lambda^*_i \le \tau^2 t^*_r$. There are at most NM_r pair–route candidates, so a Gaussian tail bound and a union bound give, with probability at least $1-\beta$,

$$\max_{u<v} \max_{P\in\mathcal{P}_r(u,v)} |\zeta_P| \le \tau\sqrt{t^*_r}\sqrt{2\ln\left(\frac{2NM_r}{\beta}\right)} =: \eta.$$

No independence is needed for this union bound. On this event, taking the minimum over routes gives

$$d^{(r)}_w(u,v) - \eta \le \min_P C_P y \le d^{(r)}_w(u,v) + \eta.$$

Since $d_w \le d^{(r)}_w \le d_w + \gamma_r(w)$ and projection onto $[0,\infty)$ is nonexpansive relative to a nonnegative target, the stated uniform error bound follows.

For feasibility, set every $\lambda_i = \Delta^2_{\mathrm{C}}$. Then $\sum_i A^2_{ie}/\lambda_i \le 1$ and every route objective is at most $\Delta^2_{\mathrm{C}} b^2_{\mathrm{C},r}$. Taking the infimum proves the dominance claim. When $r = p$, a simple shortest path uses every chord at most once, so it belongs to $\mathcal{P}_p(u,v)$ and $\gamma_p(w) = 0$.

D.3 Proof of CSR-SynGraph

Let $(\bar{X}, \bar{t})$ be a public feasible positive-definite SDP solution. For adjacent w, w' and $x = w - w'$, the edge constraints imply

$$\|\bar{X}^{1/2}Aw - \bar{X}^{1/2}Aw'\|_2 \le \sum_e |x_e| \sqrt{a_e^{\mathsf{T}} \bar{X} a_e} \le \|x\|_1 \le 1.$$

Thus the whitened release is a standard Gaussian mechanism with sensitivity one; multiplying by $\bar{X}^{-1/2}$ and all route operations preserve (ϵ_C, δ)-DP.

For each route,

$$C_P y = w(P) + \zeta_P, \qquad \mathrm{Var}(\zeta_P) = \tau^2 C_P \bar{X}^{-1} C_P^\mathsf{T} \le \tau^2 \bar{t}$$

by the Schur complement. The same NM_r-candidate Gaussian union bound as above gives

$$\max_P |\zeta_P| \le \tau\sqrt{\bar{t}}\sqrt{2\ln\left(\frac{2NM_r}{\beta}\right)}$$

with probability at least $1-\beta$. The route minimum, structural bias, and nonnegative projection then give (3).

To compare with CRH-Opt, take any feasible diagonal variance vector λ and set $X = \mathrm{diag}(1/\lambda_i)$. Its sensitivity constraints and route constraints are exactly the CSR edge and Schur-complement constraints, so every diagonal feasible point is feasible for CSR. Hence $t_r^{\mathrm{CSR}} \le t_r^*$. The isotropic point from the preceding proof gives $t_r^* \le \Delta_C^2 b_{C,r}^2$. The $r = p$ zero-bias claim follows from the existence of a simple shortest path.

For Corollary A.3, ρ-optimality gives

$$\bar{t} \le t_p^{\mathrm{CSR}} + \rho \le L\big(2(p+1)H_T L + p\big) + \rho.$$

Substitution in the CSR bound gives the displayed expression. If $p = O(1)$, $H_T = O(\log n)$, $L = O(\log n)$, $\rho = O(\log^3 n)$, and β is inverse polynomial, the variance and confidence factors multiply to $O(\log^2 n)$ up to the displayed privacy factor.

D.4 Proof of the level-aware CSR certificate

For $r = p$, every existing interval appears in its endpoint route, so every existing level is active. For each active dyadic level ℓ, assign every interval coordinate variance $\lambda_\ell = S_T/\sqrt{C_\ell}$, and assign variance one to each chord coordinate. A tree edge occurs in at most one aligned interval at each level, so its whitened squared sensitivity is at most

$$\sum_\ell \lambda_\ell^{-1} = S_T^{-1} \sum_\ell \sqrt{C_\ell} = 1.$$

The chord sensitivity is one. A candidate route uses at most C_ℓ coordinates at level ℓ and at most p chords, so its released variance is at most

$$\sum_\ell C_\ell \lambda_\ell + p = S_T^2 + p.$$

This diagonal point is feasible for the full CSR SDP. The route Gaussian tail and union bound in Theorem 3.2 prove (8); the full chord budget gives zero routing bias. Every path route uses at most two canonical dyadic intervals at each level. For a general heavy–light tree, at most $2(p+1)H_T$ coordinates occur at an active level and at most $\lceil \log_2 h_* \rceil + 1$ levels are active, giving the specializations stated in Theorem A.5.

D.5 Proof of CSR covariance transfer

Let A be the public strategy matrix and let $P_\perp$ project onto $\ker A^\mathsf{T}$. Since singleton rows are included, A has full column rank. For any $H \succ 0$ with $H_{ee} \le 1$, define

$$\Sigma_\eta = AH^{-1}A^\mathsf{T} + \eta P_\perp, \qquad \eta > 0.$$

Write $A = QR$ with Q an orthonormal basis for range(A) and R invertible. On this subspace, Σ_η has matrix $RH^{-1}R^\mathsf{T}$, and on its complement it equals ηI. Its inverse therefore gives $A^\mathsf{T}\Sigma_\eta^{-1}A = H$. For a route row C_P and $x_P = C_P A$,

$$C_P \Sigma_\eta C_P^\mathsf{T} = x_P H^{-1} x_P^\mathsf{T} + \eta \|C_P P_\perp\|_2^2.$$

The finite route family lets $\eta \downarrow 0$ uniformly. Taking $H = I$ bounds route variance by the number of tree edges and chords, at most $(p+1)D_T + p$, proving (9).

D.6 Proof of the Euler-tour edge-precision certificate

Let F map the $n-1$ tree-edge weights to the signed depth-first tour of length $2(n-1)$, recording $+w_e$ on descent and $-w_e$ on ascent. Let D contain the aligned dyadic interval indicators of this sequence, including singletons, and write $B = DF$. Every column of F has one $+1$ and one -1, and each column of D has norm at most $\sqrt{L_E}$. Therefore $\|B_{:,e}\|_2 \le 2\sqrt{L_E}$. Singleton traversal rows ensure full column rank. Set the tree-edge precision to $H_{\mathrm{tree}} = B^{\mathsf{T}}B/(4L_E)$, and set the chord block to I_p. The resulting H is positive definite with $H_{ee} \le 1$.

A prefix ending at the first visit to vertex v equals its root distance. Its canonical dyadic decoder d_v satisfies $\|d_v\|_2 \le \sqrt{L_E}$. Thus the tree-path indicator between u, v is $x_{uv} = (d_u + d_v - 2d_{\mathrm{lca}(u,v)})B =: d_{uv}B$, with $\|d_{uv}\|_2 \le 4\sqrt{L_E}$. Since $B(B^{\mathsf{T}}B)^{-1}B^{\mathsf{T}}$ is an orthogonal projector,

$$x_{uv}(B^{\mathsf{T}}B)^{-1}x_{uv}^{\mathsf{T}} \le \|d_{uv}\|_2^2.$$

For a route with at most $p+1$ tree segments, the sum of segment decoders has norm at most $4(p+1)\sqrt{L_E}$. Its variance under H^{-1} is at most $64(p+1)^2L_E^2 + p$, including independent chords. Covariance transfer proves (10) for the existing CSR strategy. For fixed p, M_p is bounded by the number of ordered oriented chord sequences, a function of p alone. The Gaussian route bound and the stated solver tolerance give Corollary A.8. Neither the auxiliary synopsis nor its decoder is released as another mechanism.

D.7 Proof of GSR-SynGraph

For adjacent w, w', let $x = w - w'$. By the definition of μ_k,

$$\|U_k^{\mathsf{T}}w - U_k^{\mathsf{T}}w'\|_2 \le \sum_e |x_e|\|U_k^{\mathsf{T}}\mathbf{e}_e\|_2 \le \mu_k\|x\|_1 \le \mu_k.$$

The coefficient release is therefore (ϵ_G, δ)-DP for $0 < \epsilon_G \le 1$. Reconstruction, clipping, and APSP are public post-processing.

Before clipping,

$$\bar{w} - w = -(I - \Pi_k)w + \eta, \qquad \eta = \sigma_k U_k g.$$

Since U_k has orthonormal columns, $\|\eta\|_2 = \sigma_k\|g\|_2$. Gaussian norm concentration gives

$$\|g\|_2 \le \sqrt{k} + \sqrt{2\ln(2/\beta)}$$

with failure probability at most $\beta/2$. Each coordinate of η has variance at most $\sigma_k^2\mu_k^2$, so a scalar Gaussian tail and a union bound over m edges give, with failure probability at most $\beta/2$,

$$\|\eta\|_\infty \le \sigma_k\mu_k\sqrt{2\ln(4m/\beta)}.$$

Coordinatewise projection onto $[0,\infty)$ cannot increase the distance to the nonnegative true vector. Thus the two events give the stated ℓ_2 and ℓ_∞ edge bounds.

For nonnegative edge vectors a, b, a simple b-shortest path has at most $H_G = n-1$ edges. Evaluating it under a gives

$$d_a(u,v) - d_b(u,v) \le \sqrt{H_G}\|a-b\|_2,$$
$$d_a(u,v) - d_b(u,v) \le H_G\|a-b\|_\infty.$$

Interchanging a, b gives absolute-value bounds, proving (4). Finally, if $w = \sum_j \alpha_j q_j$ and $\lambda_{k+1} > 0$, then

$$R_{2,k}(w)^2 = \sum_{j>k}\alpha_j^2 \le \lambda_{k+1}^{-1}\sum_{j>k}\lambda_j\alpha_j^2 \le \frac{w^{\mathsf{T}}L_{\mathcal{L}}w}{\lambda_{k+1}},$$

and $R_{\infty,k} \le R_{2,k}$.

For Corollary A.4, apply the ℓ_∞ bound, use $H_G = n-1$, $R_{\infty,k} \le \rho_k$, $\mu_k^2 \le C_\mu k/m$, and $m \ge c_m n$. This gives exactly (5); inverse-polynomial δ, β absorb the logarithms into $\widetilde{O}$. The final comparison is conditional on the stated promised class.

D.8 Proof of the constant-weight GSR specialization

If the public topology is connected, the line graph is connected and its first normalized eigenvector is $m^{-1/2}\mathbf{1}_m$. For $w = a\mathbf{1}_m$, the one-dimensional coefficient sensitivity is $1/\sqrt{m}$, so the reconstructed edge vector is $[a+Z]_+\mathbf{1}_m$ with

$$Z \sim \mathcal{N}\left(0, \frac{2\ln(1.25/\delta)}{m^2\epsilon_G^2}\right).$$

Every path distance is its hop count times a, and the released distance is the same hop count times $[a+Z]_+$. Since $a \geq 0$, $|[a+Z]_+ - a| \leq |Z|$. A one-dimensional Gaussian tail and the public hop diameter give (12).

D.9 Proof of the GSR Gaussian-process bound

Before clipping, the reconstruction error is $-r+\eta$ with $\eta = \sigma_k U_k g$. The variance of coordinate e is $\sigma_k^2\|U_k^\mathsf{T}\mathbf{e}_e\|_2^2 \leq c_{\epsilon,\delta}^2\mu_k^4$. A scalar Gaussian tail and a union bound over m edges show that

$$\|\eta\|_\infty \leq c_{\epsilon,\delta}\mu_k^2\sqrt{2\ln(4m/\beta)}$$

except with probability $\beta/2$. Under (14), this implies $w - r + \eta \geq 0$ coordinatewise, so clipping is inactive.

For the interval-prefix process, the released interval noise is controlled by the Gaussian-process comparison in the next subsection; the former pairwise union bound is unnecessary.

D.10 Gaussian-process refinement for the path theorem

Let $X_j = \mathbf{1}_{[1,j]}^\mathsf{T} U_k g$ and $X_0 = 0$. For $i < j$, orthogonal projection gives $\mathbb{E}(X_j - X_i)^2 = \|U_k^\mathsf{T}\mathbf{1}_{(i,j]}\|_2^2 \leq j - i$. Sudakov–Fernique comparison with a standard Gaussian random walk implies $\mathbb{E}\max_j X_j \leq \sqrt{2m/\pi}$ by embedding the walk at integer times in standard Brownian motion and using the reflection principle. Symmetry yields $\mathbb{E}(\max_j X_j - \min_j X_j) \leq 2\sqrt{2m/\pi}$. The range is a maximum of linear forms in g, each with coefficient norm at most $\sqrt{m}$, hence it is $\sqrt{m}$-Lipschitz. Gaussian concentration therefore bounds it by

$$\sqrt{m}\left(2\sqrt{2/\pi} + \sqrt{2\ln(2/\beta)}\right)$$

with failure probability at most $\beta/2$. Scaling by $\sigma_k \leq c_{\epsilon,\delta}\sqrt{C_\mu k/m}$ and adding the interval residual proves the stochastic term in (15). The separate edgewise event has failure probability at most $\beta/2$ and guarantees clipping is inactive under the stated margin.

D.11 Proofs for DP-FedGSR

For a client i, put $x = w_i - w_i'$ and use the public local basis U_i. Then

$$\|U_i^\mathsf{T} w_i - U_i^\mathsf{T} w_i'\|_2 \leq \sum_{e\in E_i} |x_e|\|U_i^\mathsf{T}\mathbf{e}_e\|_2 \leq \mu_i\|x\|_1 \leq \mu_i,$$

which proves Lemma A.11. The local Gaussian report is therefore (ϵ_i, δ_i)-DP for $\epsilon_i \leq 1$.

For the joint transcript, adjacent federated inputs differ only at client j. Condition on the unchanged reports Z_{-j} and apply the local DP inequality to every measurable section of a transcript event. Integrating over Z_{-j} gives

$$\begin{aligned}\Pr[M(\boldsymbol{w}) \in S] &\leq e^{\epsilon_j}\Pr[M(\boldsymbol{w}') \in S] + \delta_j \\ &\leq e^{\epsilon_{\max}}\Pr[M(\boldsymbol{w}') \in S] + \delta_{\max}.\end{aligned}$$

The reverse direction follows from symmetric adjacency. Reconstruction, clipping, disjoint union, and APSP are post-processing, proving Theorem A.12.

For edge $e \in E_i$, the pre-clipping reconstruction noise $\nu_{i,e} = \mathbf{e}_e^\mathsf{T} U_i\xi_i$ is Gaussian with variance at most $\sigma_i^2\mu_i^2$. A union bound over all m_iK local edge coordinates gives, with probability at least $1-\beta$,

$$|\nu_{i,e}| \leq \mu_i\sigma_i\sqrt{2\ln\left(\frac{2m_iK}{\beta}\right)}$$

simultaneously. The residual decomposition $U_i U_i^\top w_i - w_i = -(I - P_i)w_i$ then yields Theorem A.13. For $a \in \mathbb{R}$ and $b \geq 0$, $|[a]_+ - b| \leq |a - b|$, so clipping preserves the bound.

Finally, condition on the simultaneous edge event and let $\alpha = \|\widetilde{w} - w\|_\infty$. A simple shortest path under either nonnegative weight vector has at most H_G edges. Evaluating the true path under released weights and the released shortest path under true weights gives both directions of

$$|\widetilde{D}_{uv} - d_w(u, v)| \leq H_G \alpha.$$

Taking the maximum over pairs proves Theorem A.14; substitution of σ_i gives the displayed stochastic term and the communication count $\sum_i k_i$.

D.12 Proofs for the Proposed LEG, LDG, and LRG Methods

This section records the guarantees for the three non-spectral methods introduced in the multi-party-method subsection of the main paper. They are proposed multi-party mechanisms in this paper, rather than imported comparison algorithms. Let $E = \dot{\bigcup}_{i=1}^{K} E_i$ be the public mutually exclusive edge partition and let the protected adjacency relation change one client's vector by at most one in ℓ_1 norm.

LEG privacy and error. Client i releases $\widetilde{w}_i(e) = [w_i(e) + \eta_{i,e}]_+$ with independent $\eta_{i,e} \sim \mathrm{Lap}(1/\epsilon)$. The local vector query has ℓ_1 sensitivity at most one. Hence the Laplace mechanism is ϵ-DP for each client. Because only one client can differ under the federated adjacency relation, the joint transcript is also ϵ-DP; the disjoint union, clipping, and APSP computation are post-processing. For completeness, with probability at least $1 - \beta$, a union bound gives

$$\max_{i,e} |\eta_{i,e}| \leq \frac{\log(m/\beta)}{\epsilon}, \qquad m = |E|.$$

Clipping cannot increase the error for a nonnegative true weight. If H_G is an upper bound on the number of edges in a simple path, the standard two-sided shortest-path comparison therefore yields

$$\max_{u,v} |\widetilde{D}_{uv} - D_{uv}| \leq H_G \frac{\log(m/\beta)}{\epsilon}. \tag{20}$$

LDG privacy and structural error. For client i, let V_i be the vertices incident to its owned edges and let $f_i(w_i)$ be the vector of all finite local pair distances. A unit ℓ_1 change in edge weights changes every local shortest-path distance by at most one: evaluate a shortest path under one weight vector under the other vector, then reverse the roles. Thus, if $n_i = |V_i|$, the joint distance vector has ℓ_2 sensitivity at most n_i. Releasing

$$\widetilde{f}_i = f_i + \mathcal{N}(0, \sigma_i^2 I), \qquad \sigma_i = \frac{n_i}{\epsilon}\sqrt{2\log(1.25/\delta)},$$

is (ϵ, δ)-DP for client i when $0 < \epsilon \leq 1$. Parallel composition across the disjoint edge owners and post-processing at the server prove the joint guarantee.

Let $N = \sum_i n_i(n_i - 1)/2$ count the finite unordered local-pair releases, and let $\sigma_{\max} = \max_i \sigma_i$. A Gaussian union bound gives, with probability at least $1 - \beta$, the noise envelope

$$a_{\mathrm{LDG}} = \sigma_{\max}\sqrt{2\log(2N/\beta)}.$$

Every released shortcut represents a valid path in the union graph. After clipping its noisy weight, its error relative to the exact local distance is at most a_{LDG}, since the exact distance is nonnegative. Consequently, the weight of every shortcut is at least the corresponding global distance minus a_{LDG}. A simple route in the shortcut graph contains at most H_G edges, which gives the lower bound below. For the upper bound, every original edge is itself one of the released local-pair shortcuts, so an original global shortest path of at most H_G edges remains available with at most a_{LDG} error per edge. Thus, with probability at least $1 - \beta$,

$$\max_{u,v} |\widetilde{D}_{uv} - d_G(u, v)| \leq H_G a_{\mathrm{LDG}}. \tag{21}$$

This bound accounts for both the shortcut noise and the union topology; it does not assume a client contains the entire global shortest path.

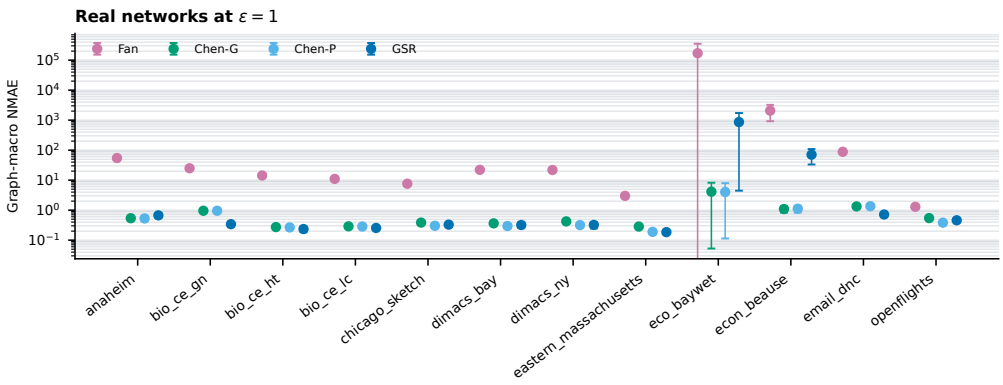


Figure 11: Centralized real-network NMAE by source at $\epsilon = 1$ for Fan, Chen Gaussian, Chen pure-DP, and GSR, averaged first over 20 noise repeats per graph and then over graphs within each source. Each source retains its own physical or constructed edge-weight unit; source-level normalization is defined in the main paper.

LRG privacy and error. Fix a public domain $[a, b]$ and write $r = (b - a)/2$, $c = (e^{\varepsilon} + 1)/(e^{\varepsilon} - 1)$. For a record $w \in [a, b]$, the proposed bounded randomized response outputs $\text{mid} \pm rc$ with probabilities chosen so that the output expectation is w. For either output atom, the ratio of probabilities under any two inputs is at most e^{ε}; this proves per-record ε-local DP. If client i protects its entire vector of m_i records, using per-record budget $\varepsilon_i = \varepsilon/m_i$ and sequential composition gives vector-level ε-LDP. Nonnegative projection and APSP are post-processing.

The released raw value differs from w by at most

$$B_i = (b - a)\frac{e^{\varepsilon_i}}{e^{\varepsilon_i} - 1},$$

and projection onto $[0, \infty)$ is nonexpansive relative to nonnegative weights. Consequently, for the union graph,

$$\max_{u,v} |\widetilde{D}_{uv} - D_{uv}| \le H_G \max_i B_i \tag{22}$$

by the same two-sided path argument as for LEG. Equations (20)–(22) establish the privacy and utility statements used in the multi-party evaluation.

E Real-Network Figures Moved from the Main Paper

The real-network plots use the fixed 136-instance collection described in the main paper. Absolute errors are not pooled across sources with different units. The multi-party plot includes DP–FedGSR, LEG, LDG, both fixed-domain LRG protocols, and the Fan/Chen local adaptations on one common comparison axis; historical LRG files with sample-dependent bounds are excluded.

Source-level heterogeneity in Fig. 11 is expected rather than a privacy-calibration effect. GSR beats Fan on all 12 sources and Chen Gaussian on 9 of 12; it beats Chen pure-DP on 5 of 12, and the losses in the other seven are mostly small margins. The two large GSR losses are *eco_baywet* and *econ_beause*, whose weights span ten and six orders of magnitude. On these graphs the global $k = 32$ basis explains only $r^2 = 0.07$ and 0.37, with residual L_∞ errors 301 and 16.6, and NMAE divides this residual by a very small mean true distance. The same two sources dominate the multi-party aggregate in Fig. 12: DP–FedGSR reaches NMAE 54.9 and 25.7 there, versus an overall graph-macro mean of 4.49. This is a basis-representation limitation on heavy-tailed weights, not a failure of the privacy mechanism; the scale and rank diagnostics isolate the same truncation term.

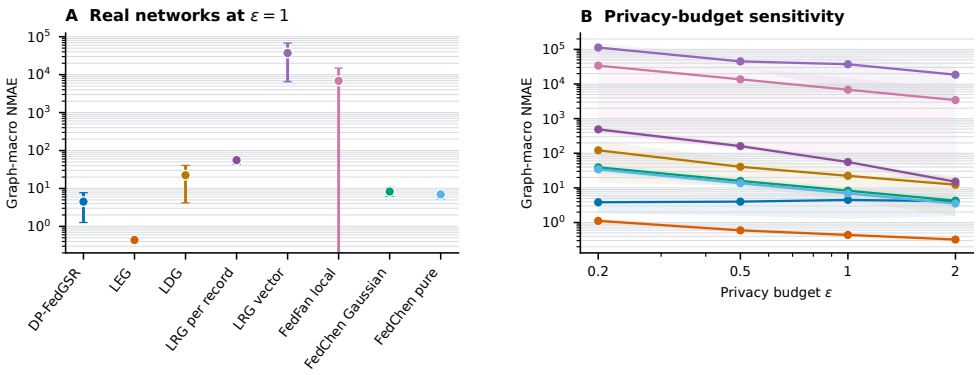


Figure 12: Real-network multi-party results. The eight plotted method blocks contain 136 graph instances and 20 repeats per graph. NMAE is averaged over repeats within each graph and then equally across graphs; error bars and ribbons show 95% confidence intervals across graph means. Colors in panel B match the methods in panel A. The FedChen rows use the repaired composition described in Section 5; LRG's per-record and client-vector rows protect different privacy units.

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